Valence is more prominent than truth conditional meaning

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What is the relative significance of Valence versus Truth-Conditional info in lexical processing?

Val First

Valence is processed first and is the most prominent feature.

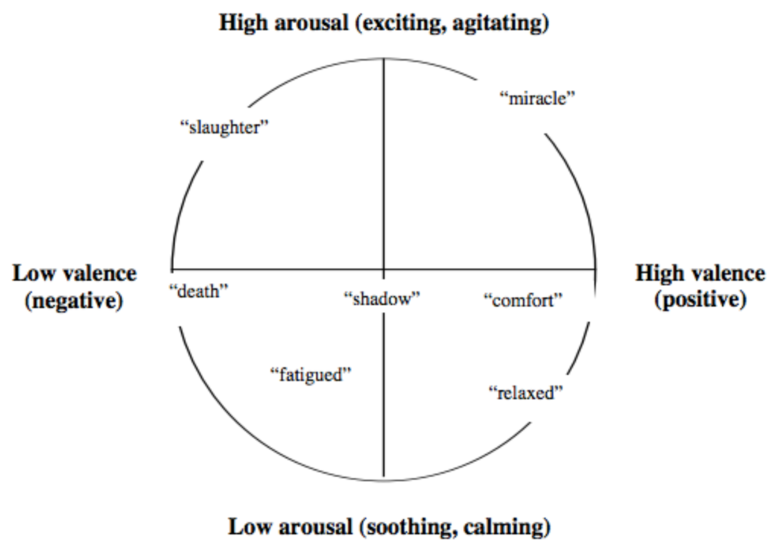
Tru-Con First

Truth-conditional meaning is processed first and is more prominent.

Val Prominence

Valence is usually more prominent than most truth-conditional features, but not necessarily all.

Affect/Valence: Ling & Psych Background

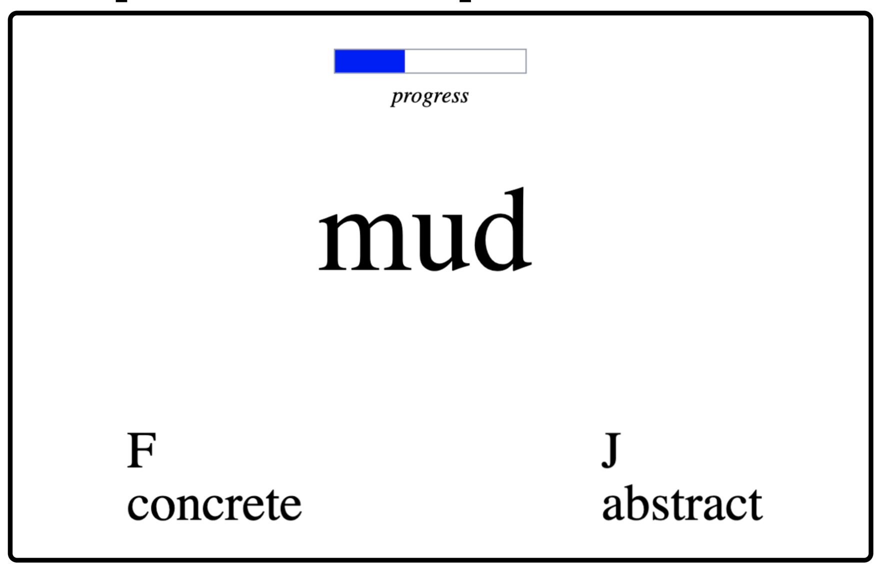


IN SEMANTICS Meaning is truth-conditions [e.g., 3-9]
IN PSYCHOLOGY Concepts located in affective space [10-16]; H1/H2 disputed empirically, for H1: [17-26], for H2: [27-35]

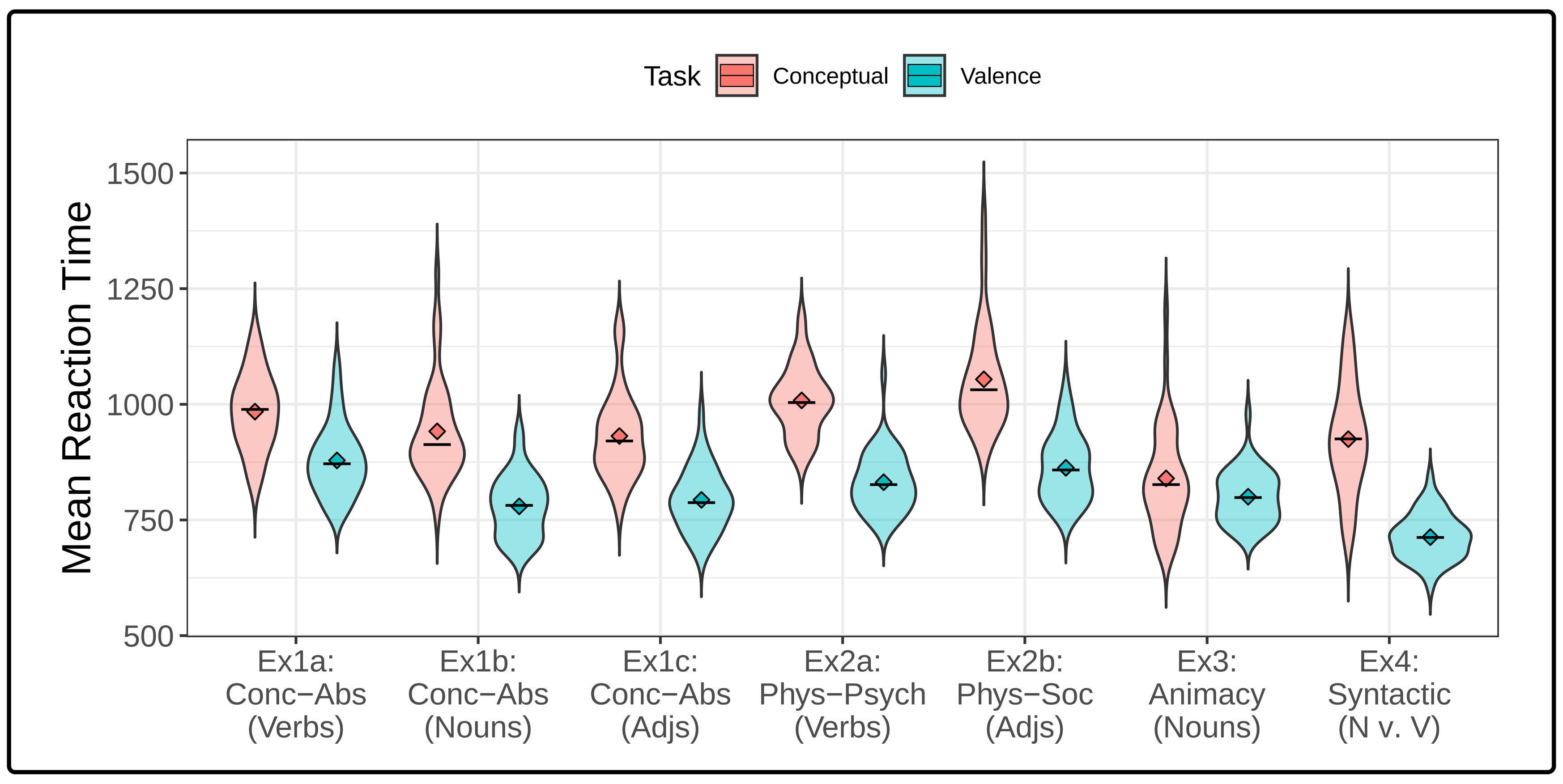
We pit Valence against many Truth-Conditional Foils:

• **Concrete/Abstract** (Exps 1a-1c) [44-54] • **Physical vs. Psychological/Social** (Exps 2a-2b) [65-70] • **Animacy** (Exp 3) [71-78] • **Syntax** (Exp 4) [e.g., 79] •

Exps 1-4: Speeded Categorization (n=270)



- **Task** 2x1 within-subjs design: **Valence**, **TruthCond**
- **DM**: forced choice
- Words normed for feature values [eg. 15,82-83]

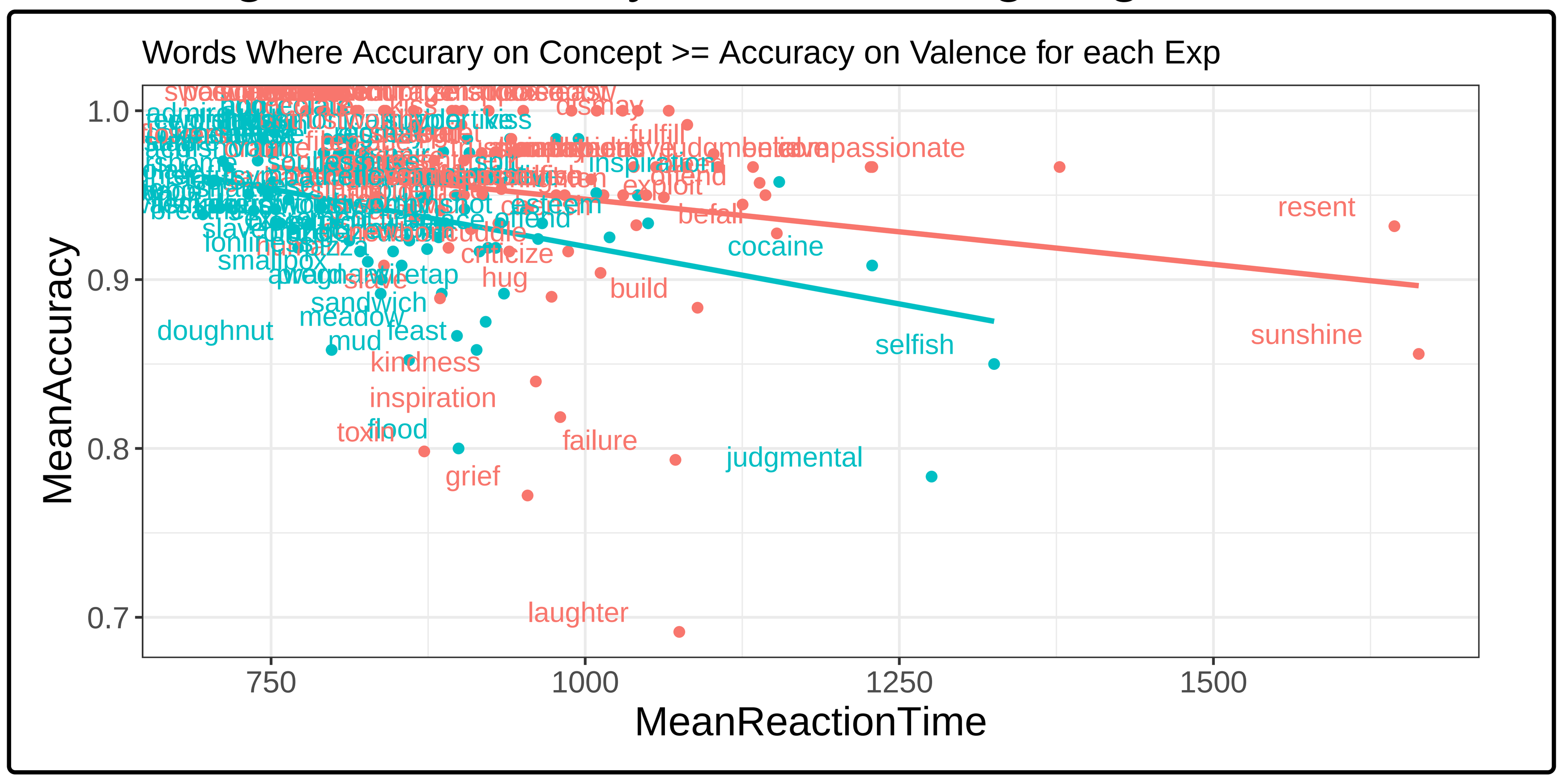


Speeded Categorization Results

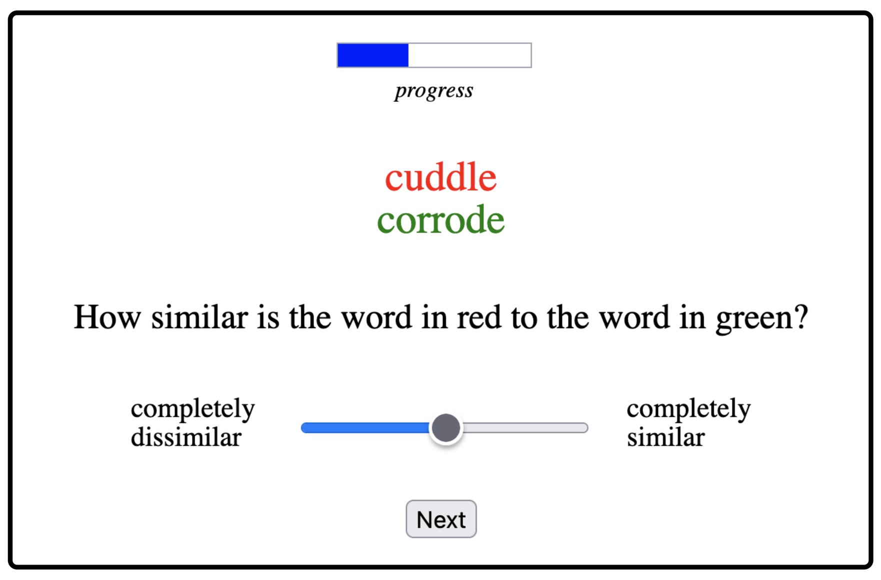
Valence < **TruthCond** except **Animacy** (Ex3)

`lmer(LogRT ~ Task + ( 1 + Task | ID) + (1 + Task | Word)`

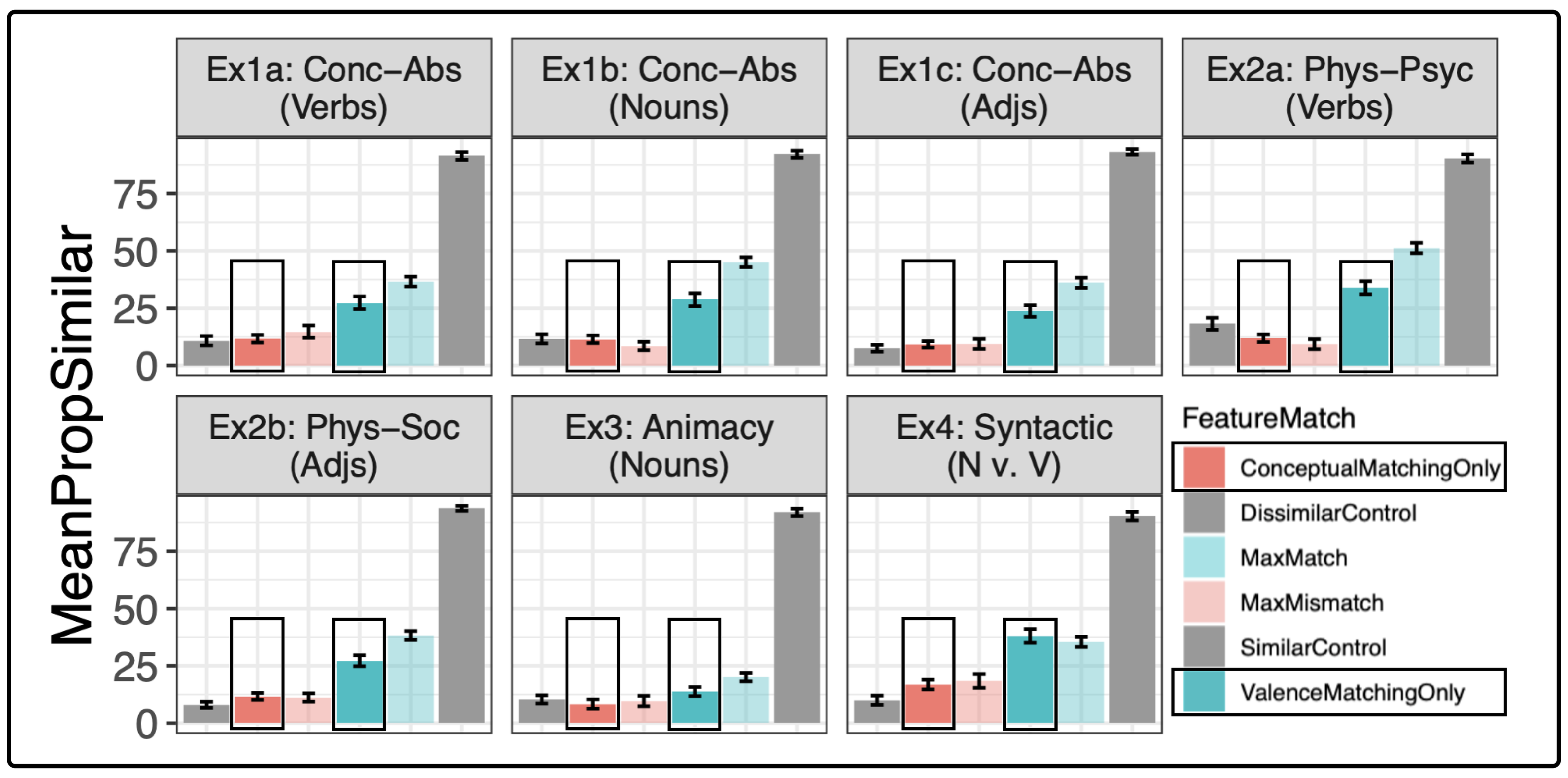
No strong RT∼Accuracy relation in high agreement words:



Exps 5-8: Similarity Judgments (n=350)



- **FeatureMatch**: 4 x 1: **ValMatch**, **TCMatch**, **MaxMismatch**, **MaxMatch**
- **DM**: slider from 1-100
- Paired words from Exps1-4

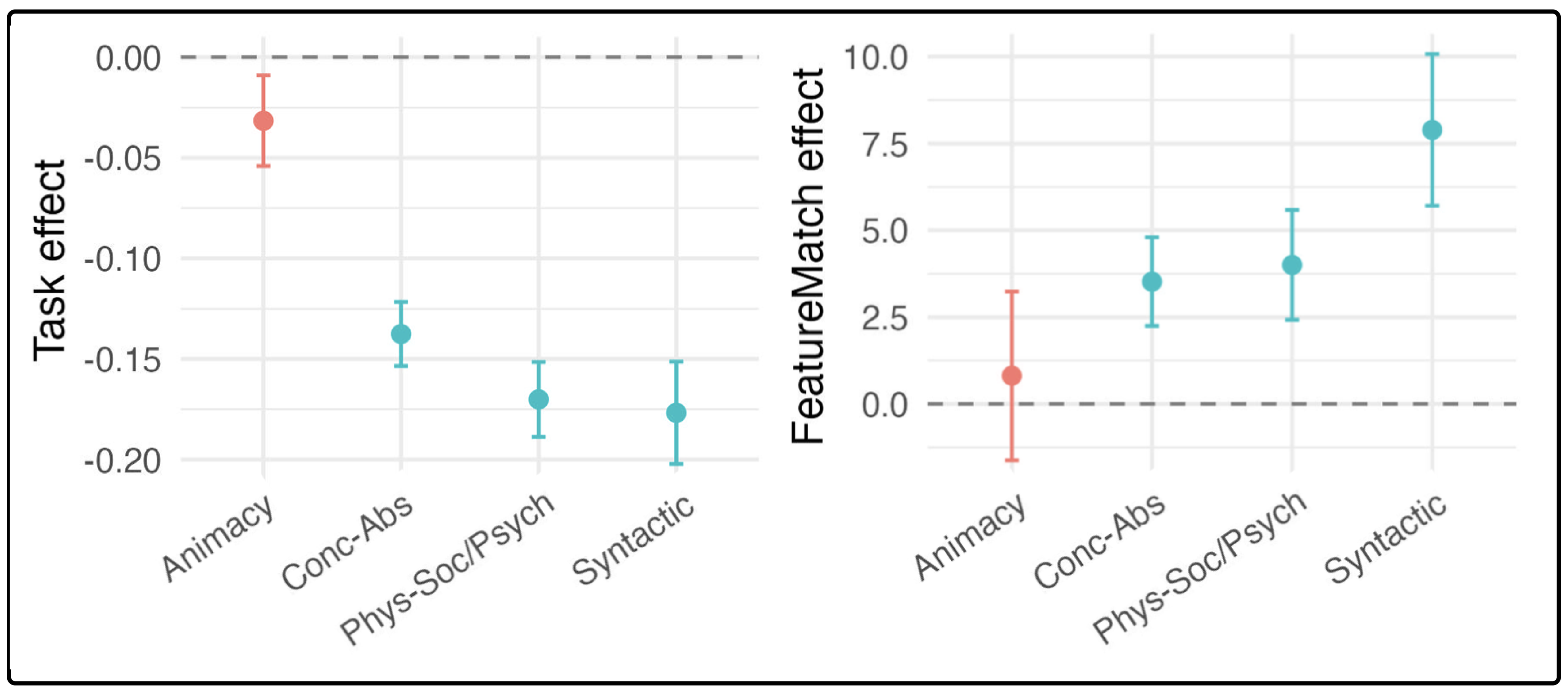


Similarity Judgment Results

ValenceMatch pairs **more similar than** **TruthCondMatch**

`lmer(Rating ~ FeatureMatch + ( 1 + FeatureMatch | ID) + (1 | Word)`

Animacy Strongest Competitor to Valence



VALENCE IS HIGHLY PROMINENT (H3)

- Participants categorized valence faster than most truth-conditional features and relied on valence more in similarity judgments.
- Pattern consistent across multiple domains: valence is accessed earlier and weighted more strongly.
- Animacy strongest non-valence competitor: not all truth-conditional features equally prominent.
- **Results support a prominence-based view in which affective info is foundational to lexical meaning.**

References

Thank you!

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